Econometrics Part 4

Heteroskedasticity and Autocorrelation

Jose Angel Garcia Sanchez

Sorbonne Data Analytics



UNIVERSITÉ PARIS 1

PANTHÉON SORBONNE

Outline

1. Generalized least squares
2. Heteroskedasticity
3. Autocorrelation

Outline

1. Generalized least squares
2. Heteroskedasticity
3. Autocorrelation

What if assumptions do not hold?

- Heteroskedasticity: error terms do not have the same variance (e.g. size effect for firms) $\rightarrow$ mostly cross sections
- Autocorrelation: the error term of period t is correlated with the error term of period $t-1$, and/or $t-2$, ... $\rightarrow$ mostly time series

In that case, $V(u) \neq \sigma^2 I_N$: OLS are still unbiased and consistent, but no longer efficient (not BLUE). Furthermore, the usual standard errors are wrong, so all the tests we derived are no longer valid. We thus have to modify the estimation to make a reliable assessment of parameter values.

The problem with OLS

Let's use the following model: $y = Xb + u$

- Assume that $V(u) = \sigma^2\Omega$, with Ω any suitable matrix (it must be symmetric and positive definite)
- $E(\hat{b}_{ols}) = b$: still unbiased
- But $V(\hat{b}_{ols}) = \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}$
- If we run an OLS regression with a software, we will get $\hat{b}$ computed as $\hat{b} = (X'X)^{-1}X'y$ (right) but $\hat{V}(\hat{b})$ computed as $\hat{V}(\hat{b}) = \hat{\sigma}^2(X'X)^{-1}$ (wrong)
- The software will thus give us unbiased estimates, but wrong variances, so all the tests we may run are wrong (T-test, F-test)

How can we correct for these problems?

- We need to transform the model so as to get homoskedastic and uncorrelated error terms
- We can use *generalized* least squares, which will replace *ordinary* least squares
- We will eventually end up re-weighting the model with the help of the variance matrix of error terms

Generalized least squares

Let's use the following model: $y = Xb + u$

- Assume that $V(u) = \sigma^2\Omega$
- Ω is symmetric and positive definite, so Ω^{-1} exists and $\Omega^{-1/2}$ exists as well
- $\Omega^{-1/2}\Omega^{-1/2} = \Omega^{-1}$
- Let's multiply the model on the left-hand side by $\Omega^{-1/2}$: $\Omega^{-1/2}y = \Omega^{-1/2}Xb + \Omega^{-1/2}u$
- The error terms of this "new" model are homoskedastic and uncorrelated: $V(\Omega^{-1/2}u) = \sigma^2\Omega^{-1/2}\Omega\Omega^{-1/2} = \sigma^2 I_N$
- We can use the transformed model for the estimation of b and most of all for inference

The GLS estimator

- Using OLS on the transformed model, we get the GLS estimator
- $\hat{b}_{gls} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$
- It is obtained by minimizing the sum of squared residuals *weighted* by Ω^{-1}
- GLS are unbiased, and BLUE (Aitken's theorem: Gauss-Markov applied to the transformed model)
- The "true" variance that can be used for tests is the following: $V(\hat{b}_{gls}) = \sigma^2(X'\Omega^{-1}X)^{-1}$
- Notice that the transformed model does not necessarily have a meaning: it is only useful for the estimation procedure

Using GLS

Since GLS amounts to OLS computed on a model that has nice properties, we can implement all the tests we described earlier (T-test, F-test).

The FGLS estimator

- GLS have nice properties
- The problem is however that we do not know the true Ω: we thus have to use an estimator of Ω, and we turn to *feasible* generalized least squares (FGLS)
- We cannot estimate all the variances and covariances of the N error terms: Ω has $N(N+1)/2$ distinct elements, and we only have N observations
- We have to impose a structure on Ω, with few parameters to estimate (see later)

Remarks

- FGLS is an estimate of true GLS
- When the sample size goes to infinity, the estimation of Ω becomes almost perfect (if Ω is consistently estimated)
- So FGLS and GLS are asymptotically equivalent
- FGLS have good properties only asymptotically
- FGLS are usually biased in small samples
- Thus, in small samples, we cannot be sure that FGLS outperform OLS

Outline

1 Generalized least squares

2 Heteroskedasticity

3 Autocorrelation

Heteroskedasticity

- Heteroskedasticity usually occurs with cross sections
- Let's assume error terms are uncorrelated but do not have the same variance (e.g. households with very different income levels)
- The variance matrix of error terms is a diagonal matrix $\sigma^2\Omega$
- It is thus easy to compute Ω^{-1}
- GLS is then sometimes called the Weighted Least Squares estimator (WLS), because each observation is weighted by the inverse of the standard deviation of its error term
- It is as if we put a smaller weight on high-income households (if their error terms have a larger variance)

Re-weighting matrix

$$\Omega = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_N \end{pmatrix} \text{ so } \Omega^{-1} = \begin{pmatrix} 1/a_1 & 0 & \cdots & 0 \\ 0 & 1/a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/a_N \end{pmatrix}$$

And the re-weighting matrix is:

$$\Omega^{-1/2} = \begin{pmatrix} 1/\sqrt{a_1} & 0 & \cdots & 0 \\ 0 & 1/\sqrt{a_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sqrt{a_N} \end{pmatrix}$$

UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Testing for heteroskedasticity

- First, simply look at the residuals (plot them against the fitted values and the X's)
- Then, various tests are available
- *Breusch-Pagan* test: regress the squared residuals on all the explanatory variables
- *White* test: regress the squared residuals on all the explanatory variables, their squares and their cross-products
- If these parameters are jointly significant (test statistic NR^2 of this regression, χ^2 with as many degrees of freedom as regressors, constant excluded), we conclude there is heteroskedasticity
- Python: `het_breuschpagan` and `het_white` from `statsmodels.stats.diagnostic`
- We will eventually use these regressions to estimate Ω and run the weighted regression

Weighted least squares in practice (1)

Let's consider a heteroskedastic model $y = Xb + u$

- The variance matrix of error terms is a diagonal with N unknown elements $V(u_i)$
- Assume that $V(u)$ depends on variables X: $V(u) = \sigma^2 \exp(X\gamma)$ (we drop i's)
- The exponential ensures the variance is always positive
- The squared residuals are the empirical counterparts of the error variances because $V(u_i) = E(u_i^2)$
- So this equation becomes empirically: $\hat{u}^2 \approx \sigma^2 \exp(X\gamma)$, so that $\log(\hat{u}^2) = X\gamma + \text{constant} + \text{error}$
- The error variances can now be estimated

Weighted least squares in practice (2)

What to do:

1. Estimate the original regression $y = Xb + u$ with OLS, save the residuals, compute their squares
2. Using plots or any other method, pick the right X's for the auxiliary regression: $\log(\hat{u}^2) = X\gamma + \text{constant}$
3. Compute the *predicted* squared residual from the auxiliary regression: $\hat{h}_i = \exp(\widehat{\log(\hat{u}_i^2)})$ (the retransformation constant does not matter here: rescaling all weights leaves the estimates unchanged)
4. For each individual i, divide every term of the original model (including the constant) by $\sqrt{\hat{h}_i}$
5. Run OLS on the transformed data, or directly `sm.WLS(y, X, weights=1/h_hat)`

Should we always use FGLS?

- FGLS do not have nice small sample properties, while OLS do
- But using OLS gives us wrong standard errors for estimates
- We could use OLS and compute corrected standard errors for them
- So we can keep OLS estimates because they are unbiased and consistent, and use the White (1980) matrix to run tests only
- That's what most people do

The White matrix

White (1980) proved that a consistent estimator for the variance matrix of parameters is:

$$\hat{V}(\hat{b}_{ols}) = (X'X)^{-1} \left(\sum_{t=1}^{N} \hat{u}_t^2 X_t X_t' \right) (X'X)^{-1}$$

Where the $\hat{u}$'s are the residuals from the OLS regression, and X_t is the (column) vector of explanatory variables of observation t.
This matrix can be used as an estimate of the true variance of OLS estimators, and thus for tests. They are called "White" or "robust" standard errors (because they are robust to heteroskedasticity). In statsmodels: `sm.OLS(y, X).fit(cov_type="HC1")` (HC0 is White's original formula; HC1 adds a small-sample correction).

Outline

1 Generalized least squares

2 Heteroskedasticity

3 Autocorrelation

Autocorrelation

- The error term of period t is correlated with the error term of period $t-1$, and/or $t-2$, ...
- Focus here on first-order autocorrelation: $u_t = \rho u_{t-1} + \varepsilon_t$
- Where $|\rho| < 1$ and ε is a "white noise"
- "White noise" means $\forall t$, $E(\varepsilon_t) = 0$ *and* $V(\varepsilon_t) = \sigma_\varepsilon^2$, the ε_t being uncorrelated with each other and with past u's

First-order autocorrelation

The variance of error terms $\sigma^2\Omega$ is no longer a diagonal matrix because Ω is no longer the identity matrix:

$$\Omega = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{N-1} \\ \rho & 1 & \rho & \cdots & \rho^{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{N-1} & \rho^{N-2} & \rho^{N-3} & \cdots & 1 \end{pmatrix}$$

And we have: $E(u_t) = 0$, $V(u_t) = \sigma^2 = \frac{\sigma_\varepsilon^2}{1-\rho^2}$.
The correlation coefficient between u_{t-h} and u_t is $\rho^{|h|}$.

Testing for first-order autocorrelation

Durbin-Watson test: $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$, with test statistic:

$$DW = \frac{\sum_{t=2}^{N} (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^{N} \hat{u}_t^2} \approx 2(1 - \hat{\rho}) \in [0, 4]$$

The Durbin-Watson table gives two values $d_L < d_U$ (which depend on N and k) such that:

$DW \in [0, d_L]$	$[d_L, d_U]$	$[d_U, 4 - d_U]$	$[4 - d_U, 4 - d_L]$	$[4 - d_L, 4]$
reject: $\rho > 0$	*inconclusive*	H_0 not rejected	*inconclusive*	reject: $\rho < 0$

Requisites: the model needs a constant term, and the lagged dependent variable can't be among the explanatory variables. statsmodels reports DW in the OLS summary.

The Breusch-Godfrey test

- This test detects autocorrelation of order higher than 1 (up to order p), and works even when the lagged dependent variable is among the explanatory variables
- Run OLS on the model
- Save the residuals e_t
- Estimate the auxiliary regression explaining e_t as a linear function of the explanatory variables $+ p$ previous residuals
- Under the null (parameters of previous residuals simultaneously equal to zero), $(N - p) \cdot R^2$ follows a χ^2 distribution with p degrees of freedom
- In the test, the R^2 used is the one from the auxiliary regression
- Python: `acorr_breusch_godfrey(res, nlags=p)` from `statsmodels.stats.diagnostic`

UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

What to do if we find autocorrelation

- Finding autocorrelation often means that the model is misspecified
- We thus should not directly switch from OLS to FGLS, but rather try to change the model
- E.g. include the lagged dependent variable (y from the previous period) as an explanatory variable, switch from x to $\log(x)$, use variables expressed as growth rates instead of levels, etc.

FGLS in the case of autocorrelation

- If modifying the model doesn't help, we have to correct for autocorrelation using FGLS, just as we did in the case of heteroskedasticity
- We will use "quasi-differencing"

Reminder: first-order autocorrelation

$$\Omega = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{N-1} \\ \rho & 1 & \rho & \cdots & \rho^{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{N-1} & \rho^{N-2} & \rho^{N-3} & \cdots & 1 \end{pmatrix}$$

So we know the form of Ω^{-1}:

$$\Omega^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho & 0 & \cdots & 0 \\ -\rho & 1+\rho^2 & -\rho & \cdots & 0 \\ 0 & -\rho & 1+\rho^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -\rho & 1 \end{pmatrix}$$

The FGLS transforming matrix

We can find a matrix Γ such that $(1-\rho^2)\Omega^{-1} = \Gamma'\Gamma$ with:

$$\Gamma = \begin{pmatrix} \sqrt{1-\rho^2} & 0 & 0 & \cdots & 0 \\ -\rho & 1 & 0 & \cdots & 0 \\ 0 & -\rho & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -\rho & 1 \end{pmatrix}$$

And the FGLS transformed model is written as: $\Gamma y = \Gamma X b + \Gamma u$.

The transformed model

- The FGLS transformed model is written as: $\Gamma y = \Gamma X b + \Gamma u$
- In the new model, y_1 is replaced by $\sqrt{1-\rho^2}\, y_1$ and all the other outcome variables y_t are replaced by $y_t - \rho y_{t-1}$
- Same for each independent variable x_j, including the constant
- Except for the 1st observation, all observations are "quasi-differenced"
- $y_t - \rho y_{t-1} = b_0(1-\rho) + b_1(x_{1,t} - \rho x_{1,t-1}) + \cdots + \varepsilon_t$
- $\forall t > 1$, $u_t - \rho u_{t-1} = \varepsilon_t$, which is a white noise: now regular OLS can be used on this new model
- The last thing we need to run the regression is an estimate of ρ, $\hat{\rho}$

Estimation: the two-step method

1. Run the original OLS regression $y = Xb + u$
2. Save the residuals $\hat{u}_t$
3. Estimate ρ by $\hat{\rho} = \frac{\sum \hat{u}_t \hat{u}_{t-1}}{\sum \hat{u}_{t-1}^2}$ (OLS regression of $\hat{u}_t$ on $\hat{u}_{t-1}$)
4. Quasi-difference the model using $\hat{\rho}$
5. Then run OLS on the transformed model

Estimation: the iterative method

Cochrane-Orcutt (drops the 1st observation) or *Prais-Winsten* (keeps it, multiplied by $\sqrt{1-\rho^2}$) method

1. Run the two-step method once, save $\hat{\rho}^{(1)}$ and $\hat{b}^{(1)}$
2. Compute the residuals of the *original* model with these new parameters: $\hat{u} = y - X\hat{b}^{(1)}$
3. Use these residuals to compute a new $\hat{\rho}$, $\hat{\rho}^{(2)}$, and quasi-difference again
4. Repeat until two successive $\hat{\rho}$'s are sufficiently close to each other

statsmodels: `sm.GLSAR(y, X, rho=1).iterative_fit()` (Cochrane-Orcutt style: drops the 1st observation)

UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Last, what if we have both autocorrelation and heteroskedasticity?

- We may use a generalization of the White variance matrix if we have both heteroskedasticity and autocorrelation that could not be corrected for
- It is used when we assume that the correlation between the u's is zero after some lag L
- We get the so-called *Newey-West standard errors*, which are heteroskedasticity and autocorrelation consistent (HAC)
- statsmodels: `sm.OLS(y, X).fit(cov_type="HAC", cov_kwds={"maxlags": L})`